Venkata Lakshmi Parimala Pasupuleti

Full Stack Developer · AI as Force Multiplier · Python · LLM Integration · RAG Pipelines · Database Design · OOP · REST APIs · AWS · CI/CD · Claude Code

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PROFESSIONAL SUMMARY

Full Stack Software Engineer with 8 years shipping production Python systems at enterprise scale - and two AI systems that show what "AI as a force multiplier" looks like in practice. At Entrata, delivered 20+ production modules across 120+ Agile sprints: OOP-based Python and JavaScript backends, database design and optimization across PostgreSQL/MySQL/Oracle (200+ queries tuned for 30% performance gain), RabbitMQ event-driven async processing, Docker containerization, and CI/CD-driven quality at 99.5% uptime for 100K+ monthly active users. Conducted 300+ code reviews and maintained sub-10 bug count for 6+ consecutive months. AgentProbe is an open-source LLM evaluation framework: 5-metric quality engine, LLM-as-judge scoring, CI-integrated regression gate, multi-provider Anthropic/OpenAI/Ollama routing - the production safety layer that ensures AI outputs are tested before they reach users. LeaseI is an 8-agent LangGraph agentic platform with full RAG pipeline, OCR document extraction, and HITL escalation - 93.6% precision across 1,240 compliance test cases. Daily Claude Code user; Anthropic API integrated throughout. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

TECHNICAL SKILLS

AI & LLM Deployment: LLM Integration, RAG Pipelines, Agentic Workflows, LangChain, LangGraph, Multi-Agent Orchestration, LLM Evaluation, LLM-as-Judge, Hallucination Detection, HITL, Guardrails, Prompt Engineering, Few-Shot Prompting, Structured Outputs, ChromaDB, Vector Search, Embeddings, Anthropic API (Claude), Claude Code, OpenAI API, Ollama, MCP, OCR, BERT, spaCy, NLP

Engineering & Backend: Python (8yr), TypeScript, JavaScript, SQL, PHP, Bash, OOP (object-oriented programming), SOLID principles, FastAPI, Flask, REST APIs, Database Design & Management, PostgreSQL, MySQL, Oracle, Redis, NoSQL, Docker, Microservices, CI/CD (GitHub Actions), Unit Testing, Debugging & Profiling, AWS (S3, EC2, RDS, Glue), RabbitMQ, Linux, Git

Reliability & Cross-Team: Scalable & Maintainable System Design, Observability, Production Monitoring, Performance Engineering, Event-Driven Systems, Data Pipelines, ETL, PHI Compliance (Libsodium), Security Best Practices, Code Review Standards, PyTorch, scikit-learn, NumPy, Pandas, Agile/Scrum

EXPERIENCE

Magizhs Tech Solution

May 2025 - Aug 2025

AI/ML Engineer Intern

- Deployed a production agentic AI system end-to-end (LangGraph, FastAPI, Python) - used Claude Code and AI coding assistants throughout the engineering lifecycle to accelerate delivery; multi-agent orchestration with RAG retrieval, deterministic guardrails, HITL escalation, and CI/CD-integrated eval gate enforcing quality at every release.
- Built observability on agent performance, retrieval latency, and system reliability; applied AI tools to accelerate code review, debugging, and test generation - demonstrating concrete productivity improvements at each sprint.

George Mason University

Aug 2024 - May 2026

Graduate Teaching Assistant & Research Associate

- Built Python ML pipelines (PyTorch, scikit-learn) on WHO and UK Road Safety datasets - $R^2 = 0.977$; end-to-end from data ingestion through feature engineering and statistical validation across 190+ countries.
- Mentored 600+ students in Python, OOP, SQL, and database design; designed observability dashboards improving research comprehension by 40%.

Entrata - PropTech / Affordable Housing SaaS

Sep 2018 - Aug 2024

Senior Software Engineer · Lehi, UT (Remote)

- **Scalable, maintainable full-stack systems and database design** - 20+ OOP-based Python and JavaScript/TypeScript production modules at 99.5% uptime; designed and optimized multi-tenant database schemas across PostgreSQL, MySQL, and Oracle; 200+ SQL query optimizations (30% performance gain via indexing, partitioning, execution plan analysis); RabbitMQ event-driven async processing with retry, idempotency, and dead-letter guarantees; Docker containerization (50% scalability improvement); AWS S3, EC2, RDS, Glue.
- **AI as force multiplier and secure system design** - AI-driven observability and monitoring tooling reducing triage time 40%; ML-powered data validation across compliance-critical workflows; 100% PHI compliance with Libsodium-encrypted pipelines (zero security incidents); Redis caching; leveraged AI development tools to accelerate debugging, code generation, and documentation across the team.
- **Code quality, unit testing, and engineering leadership** - 300+ peer code reviews enforcing OOP and SOLID design principles; unit and integration testing with CI/CD via GitHub Actions; maintained sub-10 bug count for 6+ consecutive months; mentored 5+ engineers on Python, database design, REST API patterns, and maintainable code practices; translated business requirements from product, compliance, and operations into measurable technical outcomes across 120+ Agile sprints.

Entrata India - Pune

Jan 2017 - Sep 2018

Associate Software Engineer → Trainee

- Built backend modules (PHP, Python, PostgreSQL) with OOP design patterns and database design for 1.6M HUD/Tax Credit records; AWS S3 migration resolving 90% of legacy storage constraints.

PROJECTS

AgentProbe: Open-Source LLM Evaluation & AI Dev Platform

LLM · CI/CD · Claude Code

Python · FastAPI · LangGraph · ChromaDB · CI/CD (GitHub Actions) · Anthropic · OpenAI · Ollama | github.com/venkatalakshmiparimala/AgentProbe

Production LLM evaluation platform and AI development accelerator: 15-endpoint FastAPI REST API service (OOP architecture), 5-metric quality engine (correctness, groundedness, latency, token efficiency, hallucination rate), LLM-as-judge scoring, CI-integrated regression gate, multi-provider Anthropic/OpenAI/Ollama routing. Built with Claude Code throughout - zero regressions shipped across 1,240+ test cases.

Leasel: 8-Agent Agentic Platform with Full RAG Pipeline

Agentic · RAG · HITL

Python · LangGraph (8-agent) · FastAPI · TF-IDF · OCR · SQLite · REST APIs | github.com/venkatalakshmiparimala/Lease_complaince_rag

Full-stack agentic AI system: 8-agent LangGraph stateful workflow with multi-step routing, planning, tool use, and state management; OCR document extraction and multi-source RAG retrieval; deterministic guardrails on every regulated output; HITL escalation on low-confidence decisions; source-linked answers with 100% citation correctness. 93.6% precision, 87.9% F1 across 1,240 compliance test cases.

AI on FHIR: Full-Stack NLP Enterprise Query System

Full-Stack · NLP · REST APIs

Python · Flask · React · TypeScript · Docker · BERT · spaCy · LLMs · FHIR APIs | github.com/venkatalakshmiparimala/AI_on_FHIR

Full-stack AI application (Python backend + React/TypeScript frontend) translating natural-language queries into FHIR REST API calls via BERT embeddings and LLM inference - reduced retrieval time ~60%; HIPAA-compliant RBAC with authentication, authorization, and full data isolation across structured medical records.

EDUCATION

M.S. Data Analytics Engineering - George Mason University, Virginia — May 2026 · GPA 3.87

B.Tech Computer Science & Engineering - JNTU Anantapur — 2013-2017